

Applied NLP

DATA 641

Lecture 15 — Text Summarization

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly and Natural Language Processing in Action, Understanding, analyzing, and generating text with Python, Hobson Lane, Cole Howard, Hannes Hapke, First edition, MANNING

Text Summarization

- **Text Summarization:** Creating concise summaries of longer text documents.
- **Objective:** Produce a coherent summary that encapsulates the main ideas within the text.
- **Purpose:** Facilitate quick understanding of large documents, storing only pertinent information.
- **Benefits:** Enhanced information retrieval, improved data organization, and reduced reading time.

Tasks on text summarization

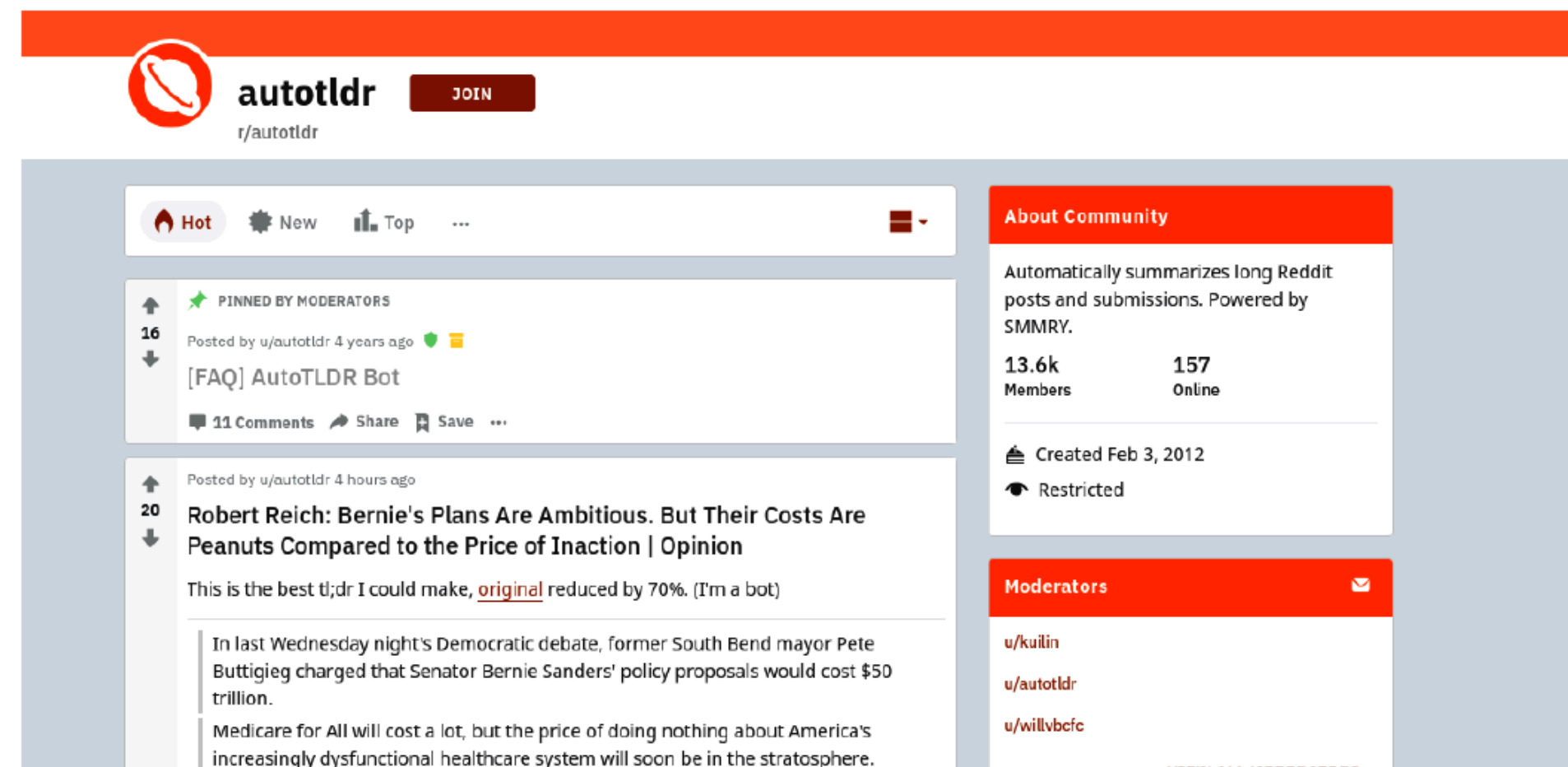
Active research in NLP on automatic text summarization began in the early 2000s. This research was a part of the Document Understanding Conference (DUC) series and DUC conferences organized competitions addressing various aspects of text summarization. Some of them include:

- **Extractive vs. Abstractive Summarization:** Extractive summarization involves selecting important sentences from a text and presenting them as a summary. Abstractive summarization focuses on generating a new summary by interpreting and rephrasing the text, rather than extracting existing sentences.
- **Query-Focused vs. Query-Independent Summarization:** Query-Focused summarization tailors the summary based on a user's query or specific information needs. Query-Independent summarization creates a general summary without considering user queries, providing a more comprehensive overview.
- **Single-Document vs. Multi-Document Summarization:** Single-Document summarization involves generating a summary from a single source document. Multi-Document summarization extends summarization to multiple documents, creating a summary from a collection of related documents.

Summarization: Use cases

Most common use case in Text Summarization:

- Single-document, query-independent, extractive summarization.
- Purpose: Create concise summaries of longer documents for human readers or machine indexing.
- Benefit: This approach enhances the user experience by providing succinct, informative summaries of lengthy content, facilitating quick understanding and efficient content indexing.



Summarization: Use cases

Two other use cases one of the authors implemented in their past workplaces are:

- An automatic sentence highlighter for news articles that colors "summary" sentences (i.e., sentences that capture the gist of the text) instead of creating full length summaries.
- A text summarizer to index only the summaries of documents instead of the full content, with the goal of reducing the size of a search engine's index.

Practical advice

There are a few practical issues to keep in mind when deploying a summarizer:

Pre-processing Importance:

- Pre-processing steps, like sentence splitting or HTML parsing, significantly influence the output summary.
- Libraries often have built-in sentence splitters, but they may not handle all cases perfectly, especially with diverse data formats.
- Custom solutions may be needed to address specific data format challenges.

Sensitivity to Input Size:

- Many summarization algorithms are sensitive to the size of the input text.
- For instance, algorithms like TextRank can be computationally expensive for larger texts.
- Solutions include summarizing large texts in partitions and combining the results or summarizing only specific portions (e.g., top $M\%$ and bottom $N\%$) if they contain the core content of the document.